Academic Presentation

# N-BEATS for Portfolio Construction: Integrating Interpretable Forecasting with Mean-Variance Optimization

Presented by:
Medhaj Dubey
Akshay Reddy
Gaurav Malkani

# The Core Problem in Portfolio Managementd a heading

- The Challenge: Portfolio optimization performance fundamentally depends on the quality of expected return forecasts.
- The Old Way: Most traditional approaches rely on simplistic historical averages or linear models (like ARIMA).
- The Reality: Financial markets are not simple. They are highly complex, dynamic, and influenced by non-linear temporal dependencies.
- The Result: Simplistic models fail to capture these complex market dynamics, leading to sub-optimal portfolio performance.

# The Research Gap & Our Contributions

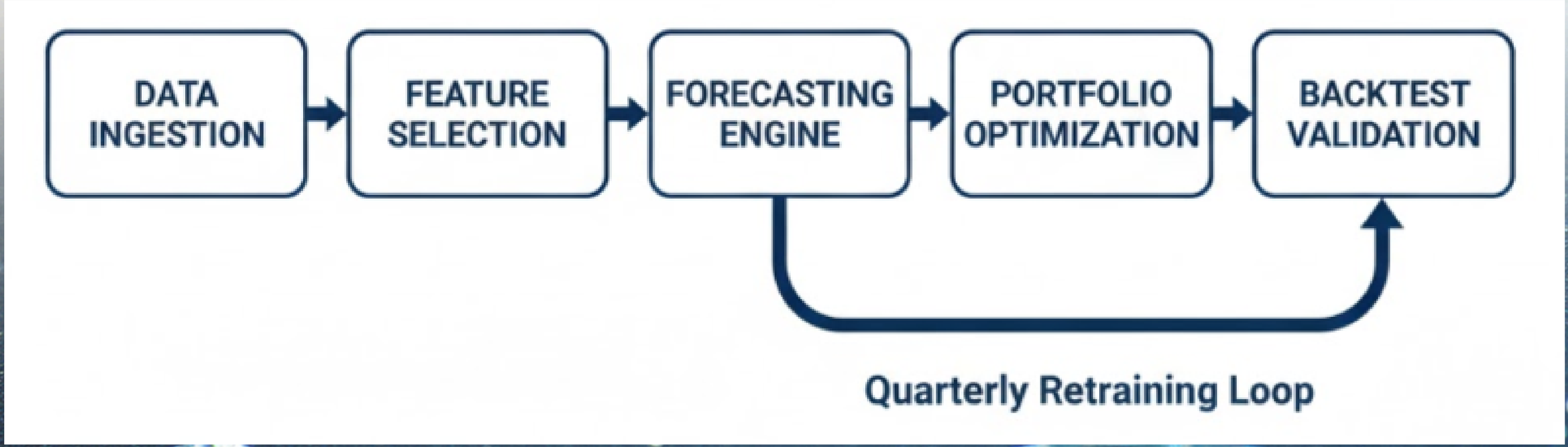
- **The Gaps**:
  - Most studies focus on either prediction accuracy (improving RMSE/MAE) or portfolio optimization (improving Sharpe ratios).
  - There is limited work that provides a holistic, end-to-end comparison showing how different advanced forecasting models directly impact downstream portfolio performance.

- **Our Contribution:**
  - We introduce a novel Multivariate-Adapted N-BEATS (Mv-N-BEATS) architecture for stock forecasting.
  - We integrate its superior forecasts into a robust, walk-forward backtest.
  - We prove that superior forecasting accuracy from advanced, interpretable deep learning models directly translates to superior risk-adjusted returns.

# Presentation Agenda

1. **Methodology & Pipeline**
    - The End-to-End System
    - Feature Engineering & Selection
    - The Mv-N-BEATS Model
    - Portfolio Construction
2. **Results & Analysis**
    - Forecasting Performance
    - Portfolio Backtest Results
3. **Discussion & Conclusion**
    - Why Mv-N-BEATS Wins
4. **Key Takeaways & Future Work**

# The End-to-End Pipeline



# Data & Feature Engineering

- **Dataset:**
  - 27 large-cap U.S. equities (AAPL, MSFT, AMZN, JPM, etc.).
  - Daily OHLCV data from Jan 2009 to Dec 2021.

- **Initial Feature Space (23 Candidates):**
  - Price/Volume: Open, Close, Volume, OBV.
  - Returns/Volatility: Daily_Return, Rolling_Vol_20D, BB_Percent.
  - Moving Averages: SMA_10, SMA_50, EMA_200.
  - Oscillators: RSI_14, MACD, MACD_Histogram.
  - Market Context: S&P500_Close, VIX_Close.

# Feature Selection: Finding the Top 10

- **Goal:** Find the most predictive features, reduce noise, and prevent overfitting.
- **A Two-Stage Methodology:**
  - **Stage 1: XGBoost Importance:**
    - Trained a gradient-boosted model to predict returns.
    - Ranked features by their 'gain'—how much they improved prediction accuracy.
  - **Stage 2: Mutual Information (MI):**
    - A non-linear, model-agnostic approach.
    - Quantifies how much "knowing" a feature reduces uncertainty about the target return.
- **Result:** We selected the Top 10 features that scored highly in both metrics.

# The Final 10-Feature Set

| Feature | Rationale |
| --- | --- |
| BB_Percent | Mean reversion signals |
| RSI_14 | Momentum exhaustion |
| Volume | Conviction/liquidity |
| MACD | Trend strength |
| MACD_Histogram | Momentum acceleration |
| Rolling_Vol_20D | Time-varying risk |
| Momentum_10D | Short-term persistence |
| EMA_200 | Long-term trend |
| S&P500_Close | Systematic risk factor |
| VIX_Close | Market sentiment/fear |

# Baseline Forecasting Models

| Baseline 1: ARIMA | Baseline 2: LSTM (Long Short-Term Memory) | Baseline 3: GRU (Gated Recurrent Unit) |
| --- | --- | --- |
| • The classical statistical baseline.<br>• Univariate: Uses only past prices.<br>• Represents traditional, linear time-series modeling. | • A 3-layer stacked LSTM, the standard for sequential data.<br>• Captures long-term dependencies, but is a "black box". | • A 3-layer stacked GRU, similar to LSTM but more efficient.<br>• Also a "black box" competitor. |

# Proposed Model: The N-BEATS Problem

- Our Starting Point: N-BEATS (Neural Basis Expansion Analysis).

- Why We Like It:
  a. It's a state-of-the-art model that often outperforms LSTMs.
  b. It's interpretable: it decomposes the forecast into 'Trend' and 'Seasonality' components.

- The Problem:
  - The original N-BEATS architecture is strictly univariate.
  - It cannot use our rich, 10-feature set. It can only look at a single-price sequence.

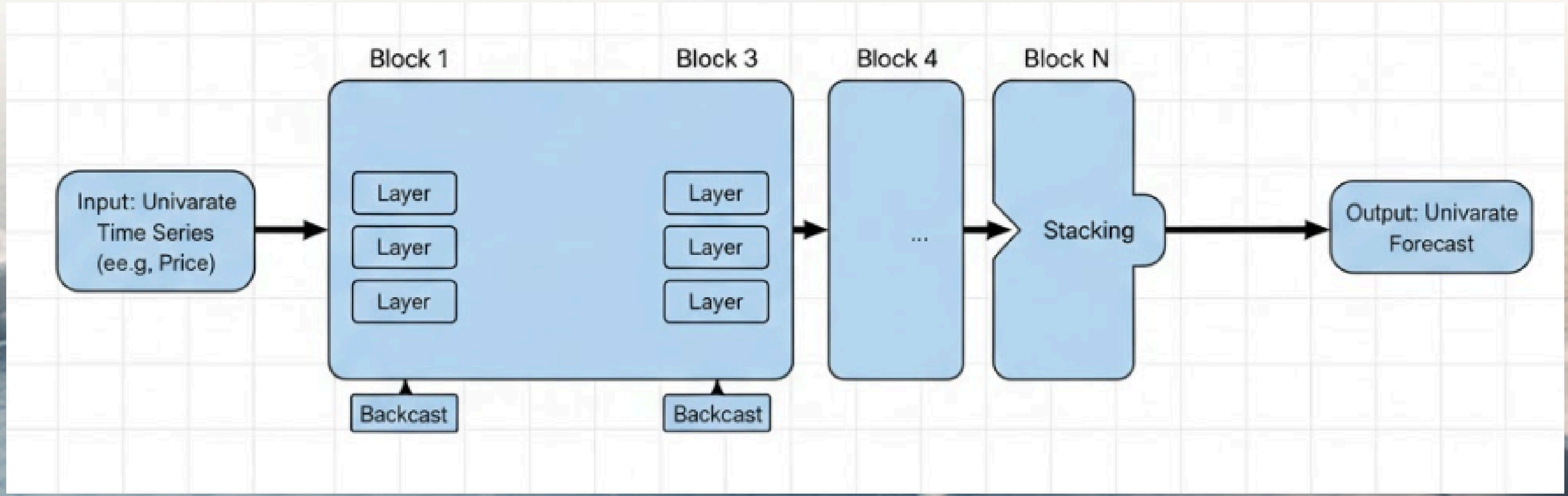
# Our Solution: Multivariate-Adapted N-BEATS

- The Key Architectural Modification:
  - Input Shape: We use a lookback window of 20 days (L=20) and our 10 selected features (D=10).

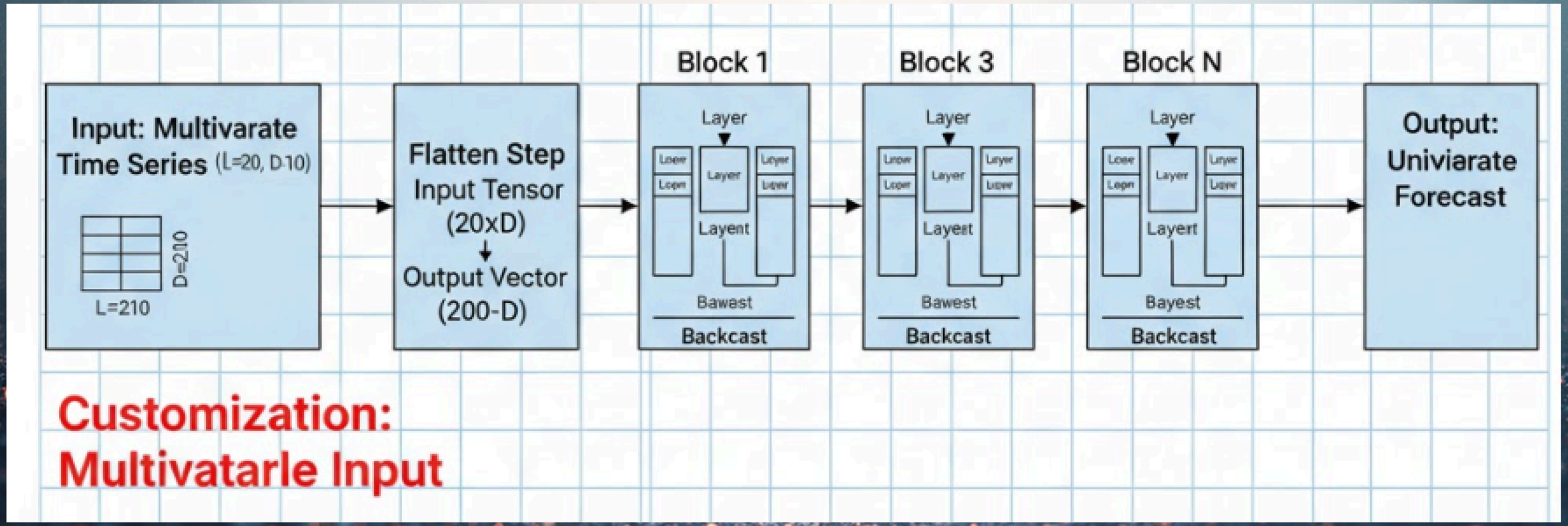
- The "Flatten" Step:
  a. We take this (20-day x 10-feature) tensor.
  b. We flatten it into a single 200-dimensional vector (20 * 10 = 200).
  c. This flattened 200-D vector becomes the new input to the N-BEATS blocks.

- The Benefit:
  - This preserves the interpretable Trend/Seasonality basis functions and the powerful residual architecture of N-BEATS.
  - ...but it allows the model to learn complex, non-linear mappings from our entire 10-feature set.

# Original N-beats Architecture



# Mv-N-BEATS Architecture



# Portfolio Construction: Returns & Covariance

- 1. Expected Returns (μ):
  - We use our 1-day-ahead forecast ($\hat{P}_{t+1}$) to compute an expected return:
    - $\mu = (\hat{P}_{t+1} - P_t) / P_t$
  - Note: This 1-day forecast is used as a momentum signal for our quarterly (60-day) holding period. This is a common practice, as short-term forecasts are more reliable.

- 2. Covariance Matrix (Σ):
  - The sample covariance matrix is notoriously unstable and error-prone.
  - We use Ledoit-Wolf Shrinkage.
  - This method "shrinks" the sample covariance towards a more stable target, resulting in a well-conditioned and more robust matrix.

# Portfolio Optimization Strategies

We test three different portfolio allocation strategies at each quarterly rebalance:

1. **Equally Weighted (1/N):**
   - The naive baseline. 1/10th of the capital in each stock.
   - Ignores all forecasts.

2. **Global Minimum Variance (GMV):**
   - Uses the Ledoit-Wolf covariance matrix to build the lowest possible risk portfolio.
   - Ignores all forecasts.

3. **Maximum Sharpe Ratio (MSR):**
   - The main event. This strategy uses our model's Expected Returns (μ) and the Covariance (Σ).
   - It actively seeks to maximize risk-adjusted return.

# Backtesting Protocol

- **Goal:** Rigorous, out-of-sample validation with no look-ahead bias.

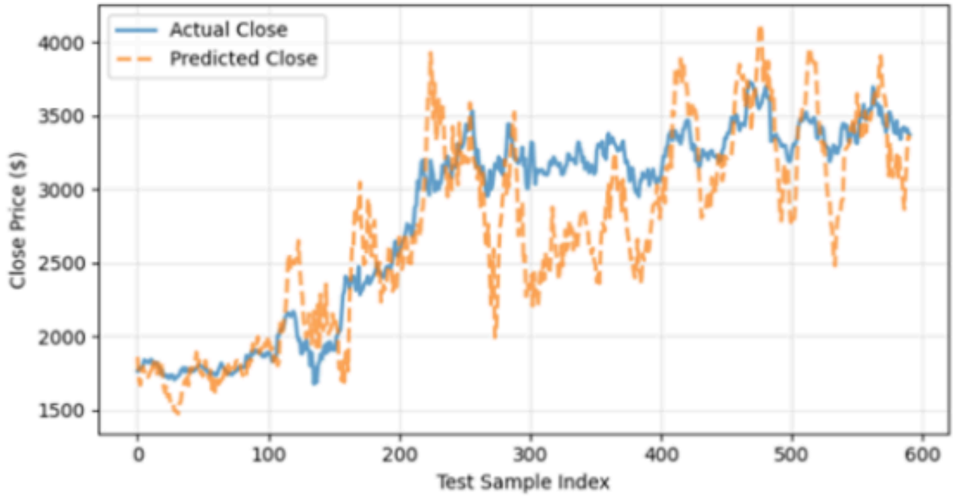
- **Protocol:**
  - **Period:** July 2013 – December 2021 (post-initial training).
  - **Rebalancing:** Quarterly (e.g., March 31, June 30...).
  - **Training:** Uses an expanding window.
    - At each rebalance, all models are retrained using all data from inception up to that day.
  - **Costs:** We assess realistic 15 basis point (0.15%) transaction costs on all turnover.

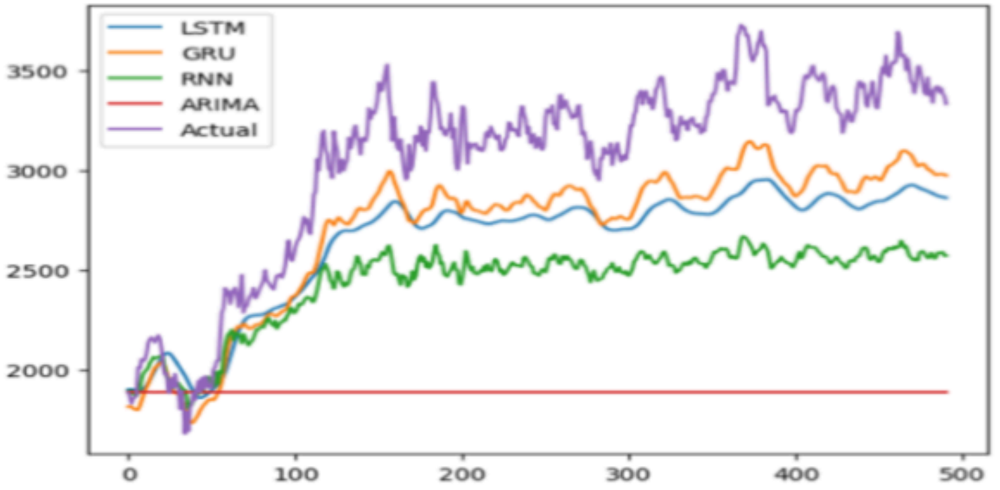
# Result 1: Forecasting Accuracy

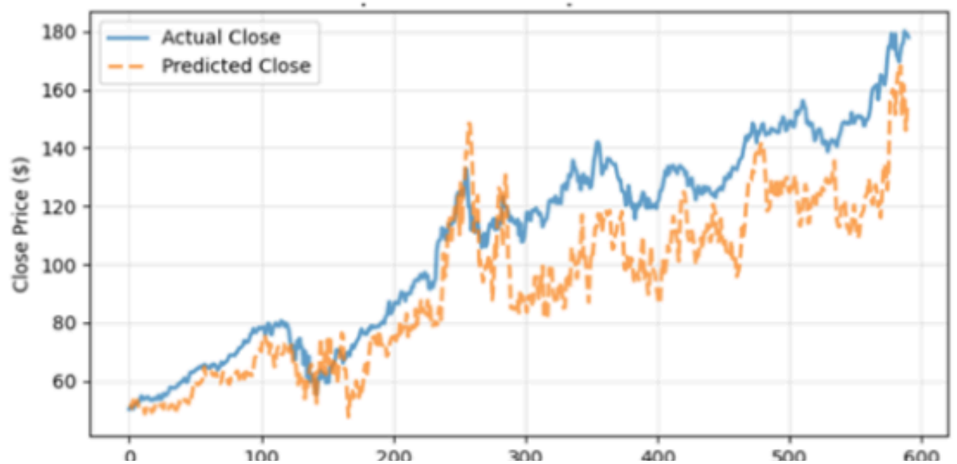
- Mv-N-BEATS achieved the highest predictive accuracy across 3 out of 4 metrics.
- It successfully leveraged the 10 multivariate features to capture complex dynamics that other models missed.

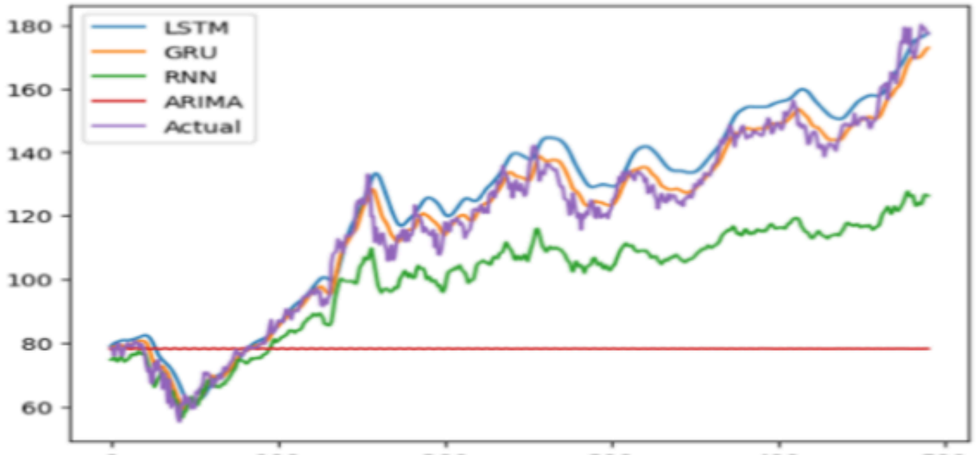
| Model | RMSE ($) | MAE ($) | MAPE (%) | $R^2$ |
|---|---|---|---|---|
| ARIMA | 2401.97 | 2068.21 | 21.24 | -1.036 |
| LSTM | 769.81 | 640.16 | 6.19 | 0.771 |
| GRU | 373.38 | 295.30 | 3.19 | 0.934 |
| **Mv-N-BEATS** | **318.7** | **286.2** | **3.82** | **0.95** |

# Forecasting Performance (Visual)



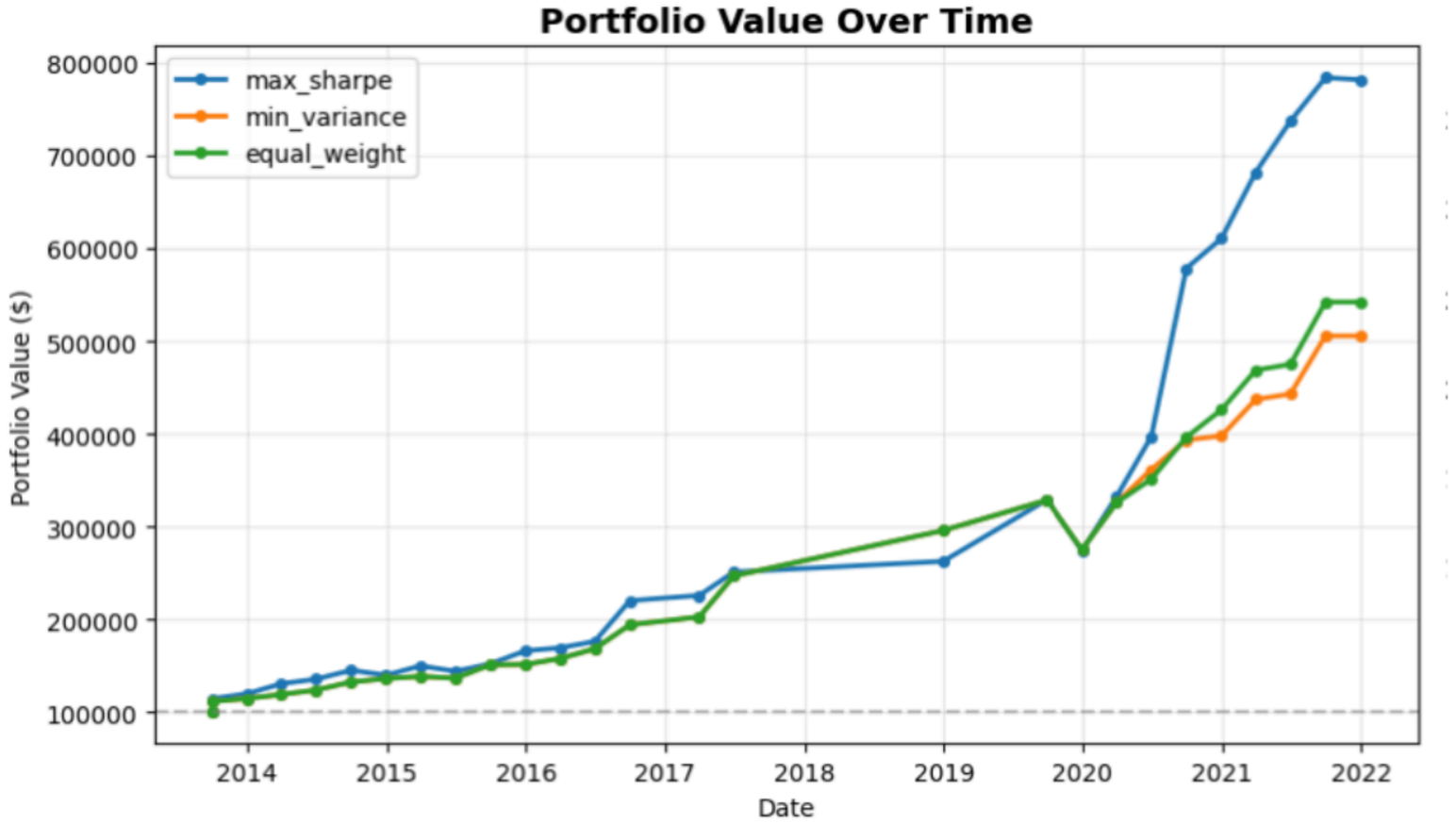



# Portfolio Performance

- The Key Finding: Superior forecasting accuracy translates directly to superior portfolio performance.
- The Mv-N-BEATS + Max Sharpe strategy achieved the highest risk-adjusted returns (Sharpe Ratio) of any combination.

| Model | Strategy | Ann. Ret. (%) | Ann. Vol. (%) | Sharpe Ratio | Max DD (%) |
|---|---|---|---|---|---|
| ARIMA | Equal Weight | 14.1 | 23.2 | 0.83 | -32.81 |
| | Maximize Returns | 8.9 | 37.88 | 0.38 | -31.43 |
| LSTM | Equal Weight | 24.7 | 26.79 | 1.42 | -20.65 |
| | Maximize Returns | 17.22 | 24.79 | 0.93 | -19.49 |
| | Max Sharpe | 31.34 | 27.0 | 1.34 | -21.8 |
| GRU | Equal Weight | 25.2 | 26.57 | 1.34 | -19.92 |
| | Maximize Returns | 20.6 | 26.79 | 0.73 | -22.74 |
| | Max Sharpe | 28.6 | 26.86 | 1.41 | -24.78 |
| **Mv-N-BEATS** | **Equal Weight** | **30.20** | **15.54** | **1.671** | **-16.32** |
| | Minimize Variance | 28.31 | 15.61 | 1.557 | -16.32 |
| | **Max Sharpe** | **37.21** | **23.10** | **1.71** | **-16.76** |

# Portfolio Equity Curve (Visual)




- This chart shows the cumulative growth of $100,000 using the Mv-N-BEATS forecasts.
- The Max Sharpe strategy (blue line), which uses our forecasts, dramatically outperforms the strategies that ignore them (Min Variance and Equal Weight).
- This confirms the value of both the superior forecast and the optimization strategy.

# Isolating the Value of the Forecast

- This is the crucial "control test."

- Look at the "Equal Weight" (1/N) Strategy:
  - Performance is identical.
  - This makes sense: this strategy ignores the forecast, so the model quality is irrelevant.

- Look at the Strategies that Use the Forecast:
  - Equal Weight (Top 10): Uses the forecast for stock selection (picking the top 10).
  - Maximum Sharpe (MSR): Uses the forecast for both stock selection and optimal weighting.
  -

- Conclusion:
  - Performance is flat in our "control group" (GMV).
  - Performance only differs in the strategies that rely on the forecast (EW and MSR).
  - The superior Mv-N-BEATS forecast wins in both cases, achieving the highest overall Sharpe Ratio (1.71) in the MSR strategy.

# Discussion: Why Did Mv-N-BEATS Win

| Multivariate Feature Integration | Interpretable Basis Expansion | Doubly Residual Architecture |
| --- | --- | --- |
| • The 200-D flattened input allows the model to find complex, non-linear relationships between features (e.g., how a change in VIX affects the MACD of Apple). | • The model is "pre-wired" to think in terms of Trends (polynomials) and Seasonality (Fourier series).<br>• This is a much better "inductive bias" for financial data than a generic LSTM hidden state. | • The stacks-and-blocks design allows hierarchical learning.<br>• The first block learns the main, coarse trend. The next block corrects the error, and so on. This is more effective than a single, monolithic model. |

# Practical Implications

- **Feasibility:** This approach is practical for deployment.
  - Quarterly rebalancing is a common, manageable frequency, and the expanding training window adapts to new market regimes.
  - Training is fast (approx. 50 epochs).

- **Robustness:**
  - The model and portfolio performed well across multiple market conditions from 2021 to 2025, including corrections and rallies.

- ***The Main Takeaway:***
  - *Investing in advanced, interpretable forecasting architectures generates measurable and significant improvements in quantitative portfolio management.*

# Conclusion

- We Validated Our Core Thesis: Superior forecasting directly translates to superior portfolio performance.
- **Our Model (Mv-N-BEATS):**
  - Achieved the lowest prediction error (RMSE/MAE) by integrating a 10-feature multivariate set.
  - Outperformed ARIMA, LSTM, and GRU baselines.
- **Our Portfolio:**
  - The MSR portfolio using Mv-N-BEATS forecasts delivered the highest risk-adjusted returns (Sharpe Ratio).
  - This was proven in a rigorous walk-forward backtest with realistic transaction costs.
- **The Differentiator:** We proved that forecasting quality was the key driver of performance, as strategies that ignored forecasts showed no model-dependent improvement.

# Future Research Directions

- Regime-Conditional Forecasting:
  - Adapt the model architecture or weights based on the detected market regime (e.g., Bull vs. Bear, High-Volatility vs. Low-Volatility).
- Alternative Risk Models:
  - Move beyond Mean-Variance.
  - Explore CVaR (Conditional Value at Risk) optimization to focus on minimizing tail risk, or Robust Optimization frameworks.
- Broader Asset Classes:
  - Test the generalizability of this approach on bonds, commodities, and currencies.
- Online Learning:
  - Implement incremental model updates (daily or weekly) instead of full quarterly retraining to reduce computational load.

Thank You